CHURNGUARD

Predict churn.
Prioritize retention.

An executive review of customer risk, model performance, revenue exposure and actionable retention strategy.

CUSTOMER CHURN PREDICTION | BUSINESS PRESENTATION

Executive summary

The decision: focus retention resources where modelled exposure is concentrated

94% recall

XGBoost identifies most churners in the held-out test set.

372 priority accounts

High and Critical risk accounts represent 16.52% of the evaluated sample.

78.60% of exposure

Share of estimated probability-weighted monthly revenue at risk.

BUSINESS IMPLICATION

Start with targeted outreach, not blanket discounting.

Pilot retention offers and validate incremental benefit before scaling.

From data to decision

Two-row workflow with deployment alongside evaluation and strategy

1. Understand

Customer data

2. Explore

Patterns & quality

3. Prepare

Clean & encode

4. Develop

Train models

5. Evaluate

Performance

6. Segment

Churn risk

7. Strategize

Retention & ROI

8. Streamlit

Single / batch predictions

What the customer data tells us

Observed customer mix and data-readiness findings

Mobile-first base

66.4% use mobile as their login device. Mobile engagement should be a priority.

Card-led payments

Debit cards (40.7%) and credit cards (31.2%) are the most common payment methods.

Segment concentration

Super (36.1%) and Regular Plus (34.3%) form the largest account segments.

DATA QUALITY

Categorical missingness is low (0.9%-2.0%), but labels require standardization

and missing values need imputation before reliable scoring.

Model performance

Tuned XGBoost selected for strong churn detection

96%	86%	94%	0.90	0.96
Accuracy	Precision	Recall	F1-score	PR-AUC

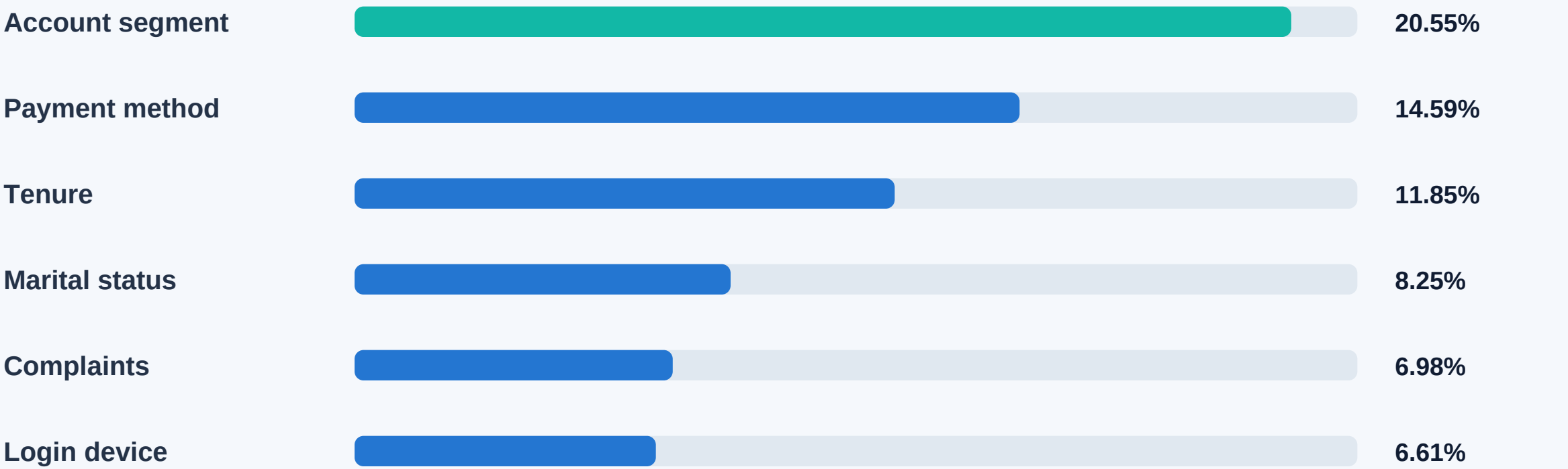
MODEL SELECTION

Cross-validation PR-AUC: 0.9481 | Test ROC-AUC: 0.99

High recall supports early intervention; precision informs outreach cost.

What drives the predictions

Top feature contributions from the fitted model



Interpretation: feature importance measures predictive contribution, not causation or risk direction.

Revenue exposure is highly concentrated

Risk scoring provides a practical starting point for prioritization

372

High + Critical risk accounts

16.52% of accounts | 78.60% of modelled monthly revenue exposure

Priority 1: High / Critical

Focus personalized interventions on the accounts with the largest expected revenue exposure.

Priority 2: Moderate / Low

Use lower-cost monitoring and preventive engagement rather than broad incentives.

Exposure is probability-weighted, model-based and expressed in the dataset's original revenue units.

Retention strategy & illustrative ROI

Test low-cost actions before committing to expensive incentives

1. Personalized outreach

5% assumed offer cost. Highest illustrative ROI across modelled scenarios.

2. Loyalty rewards

10% assumed offer cost. Candidate for a targeted controlled pilot.

3. Discount vouchers

15% assumed offer cost. Use selectively for priority accounts.

SPENDING GUARDRAIL

Premium benefits (60%) show negative ROI in the Conservative scenario.

All ROI results are scenario estimates, not measured campaign outcomes.

Decision and next steps

A controlled pilot turns a strong model into measurable retention value

01

Deploy scoring

Use Streamlit for individual and batch scoring, risk categories and CSV outputs.

02

Pilot interventions

Randomize targeted outreach, loyalty rewards and vouchers against control groups.

03

Measure impact

Track retention uplift, incremental net benefit, real costs and calibration.

04

Scale selectively

Expand only where measured outcomes justify spending.

Caveat: no churn timestamps, observed campaign outcomes, verified costs, margins or defined revenue units.